



WORK SHEET - 33

DIFFERENTIAL EQUATION + TRIGONOMETRY

Q.1 (B)

$$\frac{dP}{dx} = (100 - 12\sqrt{x}) \quad (\text{variable separable})$$

$$\therefore \int dP = \int (100 - 12\sqrt{x}) dx$$

$$\Rightarrow P(x) = 100x - 8x\sqrt{x} + c$$

$$\text{As, } P(x=0) = 2000 \Rightarrow c = 2000$$

$$\therefore P(x) = 100x - 8x\sqrt{x} + 2000$$

$$\text{So, } P(x=25) = 2500 - 1000 + 2000 = 3500 \quad \text{Ans.}$$

Q.2 (C)

$$\frac{dP(t)}{dt} + \left(\frac{-1}{2}\right)P(t) = -450 \quad (\text{Linear differential equation})$$

$$\therefore \text{I.F.} = e^{\frac{-t}{2}}$$

Also, general solution is

$$P(t)e^{\frac{-t}{2}} = 900e^{\frac{-t}{2}} + c$$

$$\text{As, } P(0) = 850 \Rightarrow c = -50$$

$$\therefore P(t) = 900 - 50e^{\frac{t}{2}}$$

$$\text{Let } t = T, \text{ then } P(t=T) = 0$$

$$\Rightarrow 50e^{\frac{T}{2}} = 900$$

$$\Rightarrow T = 2 \ln 18 \quad \text{Ans.}$$

Q.3 (A)

$$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} \quad \dots\dots(1)$$

$$\text{Also, } \frac{-dV}{dt} = 72\pi \quad \dots\dots(2)$$

$$\therefore \text{From (1) and (2), we get } -72\pi = 4\pi r^2 \frac{dr}{dt}$$

$$\Rightarrow \int r^2 dr = -18 \int dt \Rightarrow \frac{r^3}{3} = -18t + c \quad \dots\dots(3)$$

$$\text{Also, at } t=0, \quad \frac{4}{3}\pi r^3 = 4500\pi \quad (\text{Given})$$

$$\Rightarrow r^3 = 3375$$

$$\text{So, put } t=0 \text{ in equation (3), we get } c = 1125$$

$$\Rightarrow r^3 = -54t + 3375$$



∴ Differentiate both sides with respect to t, we get

$$3r^2 \frac{dr}{dt} = -54 \Rightarrow \left. \frac{dr}{dt} \right|_{r=9} = \frac{-18}{r^2} = \frac{-18}{81} = \frac{-2}{9} \quad \text{Ans.}$$

[when $t = 49$, $r^3 = 729 \Rightarrow r = 9$]

Q.4 (D)

$$\int \frac{dy}{y+3} = \int dx \quad (\text{variable separable})$$

$$\Rightarrow \ln(y+3) = x + c$$

$$\text{As, } y(0) = 2 \Rightarrow c = \ln 5$$

$$\therefore y = -3 + 5e^x$$

$$\text{So, } y(\ln 2) = -3 + 10 = 7 \quad \text{Ans.}$$

Q.5 (B)

$$\frac{dx}{dy} = \frac{e^{\tan^{-1}y} - x}{1+y^2}$$

$$\frac{dx}{dy} + \left(\frac{1}{1+y^2} \right) x = \frac{e^{\tan^{-1}y}}{1+y^2} \quad (\text{linear differentiable equation})$$

$$\therefore \text{I.F.} = e^{\int \frac{dy}{1+y^2}} = e^{\tan^{-1}y}$$

$$\text{So, general solution is } x \cdot e^{\tan^{-1}y} = \int \left(\frac{e^{\tan^{-1}y}}{1+y^2} \right) \cdot e^{\tan^{-1}y} dy$$

$$\text{Put } e^{\tan^{-1}y} = t \Rightarrow x \cdot e^{\tan^{-1}y} = \frac{e^{2\tan^{-1}y}}{2} + c$$

$$\Rightarrow 2x \cdot e^{\tan^{-1}y} = e^{2\tan^{-1}y} + k \quad \text{Ans.}$$

Q.6 (C)

$$x^2 + y^2 = 2ay \quad \dots\dots(1)$$

Differentiate both sides with respect to x, we get

$$2x + 2y \frac{dy}{dx} = 2a \frac{dy}{dx} \Rightarrow x + (y-a) \frac{dy}{dx} = 0 \quad \dots\dots(2)$$

$$\text{Put } a = \left(\frac{x^2 + y^2}{2y} \right) \text{ from (1) in (2), we get}$$

$$x + \left(y - \frac{(x^2 + y^2)}{2y} \right) \frac{dy}{dx} = 0 \Rightarrow x + \frac{(y^2 - x^2)}{2y} \frac{dy}{dx} = 0 \Rightarrow 2xy - (x^2 - y^2) \frac{dy}{dx} = 0$$

$$\therefore (x^2 - y^2)y' = 2xy$$

Q.7 (B)

$$(ydx + xdy) + x^2ydy = 0$$

$$\Rightarrow d(xy) = -\frac{x^2y^2}{y} \cdot dy \quad (\text{exact form})$$

$$\Rightarrow \int \frac{d(xy)}{(xy)^2} + \int \frac{dy}{y} = 0 \Rightarrow \frac{-1}{xy} + \ln y = c \quad \text{Ans.}$$

Q.8 (D)

$$Ax^2 + By^2 = 1 \quad \dots\dots(1)$$

$\therefore$ Differentiate w.r.t x, we get

$$2Ax + 2By \frac{dy}{dx} = 0 \quad \dots\dots(2)$$

Again differentiate w.r.t. x, we get

$$A + B((y')^2 + yy'') = 0$$

$$\Rightarrow B((y')^2 + yy'') = -A \quad \dots\dots(3)$$

$\therefore$ Using (3) in (2), we get

$$-B((y')^2 + yy'')x + Byy' = 0$$

$$\Rightarrow -\left(\left(\frac{dy}{dx}\right)^2 + y \frac{d^2y}{dx^2}\right)x + y \frac{dy}{dx} = 0$$

$\therefore$ The above differential equation is of second order and first degree.

Q.9 (C)

$$\text{We have, } (x-a)^2 + y^2 = a^2$$

$$\Rightarrow x^2 + y^2 - 2ax = 0, \quad \dots\dots(1) \quad \text{where } a \text{ is parameter}$$

$\therefore$ Differentiate both sides with respect to x, we get

$$2x + 2y \frac{dy}{dx} = 2a \Rightarrow x + y \frac{dy}{dx} = \frac{x^2 + y^2}{2x} \Rightarrow 2x^2 + 2xy \frac{dy}{dx} = x^2 + y^2 \Rightarrow y^2 = x^2 + 2xy \frac{dy}{dx}$$

Q.10 (B)

$$2xy \frac{dy}{dx} = (x^2 + y^2) \Rightarrow (2xydy - y^2dx) = x^2dx$$

$$\Rightarrow \left(\frac{2xydy - y^2dx}{x^2} \right) = dx \Rightarrow \int d\left(\frac{y^2}{x} \right) = \int dx \Rightarrow \frac{y^2}{x} = x + c$$

As, P(2, 1) satisfy it, so

$$\Rightarrow \frac{1}{2} = 2 + c \Rightarrow c = \frac{-3}{2} \Rightarrow y^2 = x^2 - \frac{3}{2}x \Rightarrow x^2 - \frac{3}{2}x - y^2 = 0$$

$$\Rightarrow \frac{\left(x - \frac{3}{4}\right)^2}{1} - \frac{y^2}{1} = \frac{9}{16}, \text{ which is equation of rectangular hyperbola, whose eccentricity} = \sqrt{2}.$$

Q.11 (B)

$$\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2xy} \Rightarrow (2xy \, dy - y^2 dx) = (x^2 + 1) dx$$

$$\Rightarrow \frac{(2xy \, dy - y^2 dx)}{x^2} = \left(1 + \frac{1}{x^2}\right) dx \Rightarrow \int d\left(\frac{y^2}{x}\right) = \int d\left(x - \frac{1}{x}\right)$$

$$\therefore \frac{y^2}{x} = x - \frac{1}{x} + c \Rightarrow y^2 + 1 = x^2 + cx \quad \text{Ans.}$$

Q.12 (A)

$$\frac{dy}{dx} = \sqrt{y-x}$$

$$\Rightarrow \frac{dy}{dx} - 1 = (\sqrt{y-x} - 1) = t \quad (\text{Subtract 1 on both sides})$$

$$\Rightarrow \int \frac{d(y-x)}{\sqrt{y-x}-1} = \int dx$$

$$\text{Put } (\sqrt{y-x} - 1) = t \Rightarrow \frac{1}{2\sqrt{y-x}} d(y-x) = dt \Rightarrow d(y-x) = 2(1+t)dt$$

$\therefore$ we get

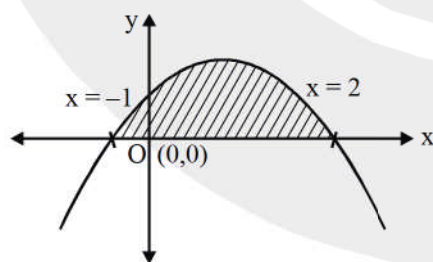
$$2 \int \frac{(t+1)dt}{t} = \int dx \Rightarrow 2(t + \ln|t|) = x + c \Rightarrow x = 2(\sqrt{y-x} + \ln|\sqrt{y-x}-1|) + k \quad \text{Ans.}$$

Aliter :

Put $y-x = t$ and do separation of variables.

Q.13 (A)

$$\text{Let } I = \int_a^b (2+x-x^2) dx$$



Graph of $f(x) = 2 + x - x^2$

$\therefore$ Clearly from above graph $\int_a^b (2+x-x^2) dx$ will be maximum when $a = -1, b = 2$

Q.14 (B)

Given, $\cos 2x > |\sin x|$, $x \in \left(-\frac{\pi}{2}, \pi\right)$

Now, we shall consider two cases :

Case-I : $x \in (0, \pi)$, we get $\cos 2x > \sin x$

$$\Rightarrow 1 - 2 \sin^2 x > \sin x \Rightarrow (2 \sin x - 1)(\sin x + 1) < 0 \Rightarrow -1 < \sin x < \frac{1}{2}$$

$$\therefore x \in \left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, \pi\right) \dots\dots(1)$$

Case-II : $x \in \left(-\frac{\pi}{2}, 0\right]$, we get $\cos 2x > -\sin x$

$$\Rightarrow 1 - 2 \sin^2 x + \sin x > 0 \Rightarrow (2 \sin x + 1)(\sin x - 1) < 0 \Rightarrow -\frac{1}{2} < \sin x < 1$$

$$\therefore x \in \left(-\frac{\pi}{6}, 0\right]$$

$\therefore$ From (1) $\cap$ (2), we conclude $x \in \left(-\frac{\pi}{6}, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, \pi\right)$

Q.15 (B)

Let $AA_1 = d_1$; $BB_1 = d_2$

In rt. triangle AGB_1 , $\frac{4}{9}d_1^2 + \frac{d_2^2}{9} = \frac{9}{4} \dots\dots(1)$

Similarly, in rt. triangle BGA_1 , $\frac{4}{9}d_2^2 + \frac{d_1^2}{9} = 4 \dots\dots(2)$

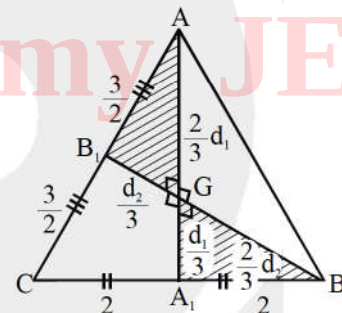
$\therefore$ From (1) and (2), we get

$$d_1 = \sqrt{3}, d_2 = \frac{\sqrt{33}}{2}$$

Now, $\text{ar}(\Delta ABC) = 3 \text{ ar}(\Delta GAB)$

$$= 3 \left(\frac{1}{2} \times \left(\frac{2d_1}{3} \right) \times \left(\frac{2d_2}{3} \right) \right) = \left(\frac{2}{3} \right) d_1 d_2 = \frac{2}{3} \frac{(\sqrt{3})(\sqrt{3}\sqrt{11})}{2} \Rightarrow \Delta = \sqrt{11}$$

Ans.



Q.16 (B)

Given, In ΔABC

$$a^2 + c^2 = 2002b^2$$

$$\text{Now, } \frac{\cot A + \cot C}{\cot B} = \frac{\left(\frac{\cos A}{\sin A} \right) + \left(\frac{\cos C}{\sin C} \right)}{\left(\frac{\cos B}{\sin B} \right)} = \frac{\sin(A+C) \cdot \sin B}{\sin A \cdot \sin C \cdot \cos B} = \frac{\sin^2 B}{\sin A \cdot \sin C \cdot \cos B}$$

$$= \frac{b^2}{(ac) \cdot \cos B} = \frac{b^2}{(ac) \left(\frac{a^2 + c^2 - b^2}{2ac} \right)} = \frac{2b^2}{a^2 + c^2 - b^2} = \frac{2b^2}{2001b^2} = \frac{2}{2001} \text{ Ans.}$$

Q.17 (D)

In $\triangle ABC$, given

$$\sin A + \sin B + \sin C = 1 + \sqrt{2} \quad \dots\dots(1)$$

$$\cos A + \cos B + \cos C = \sqrt{2} \quad \dots\dots(2)$$

$\therefore$ Equation (1) and equation (2) are satisfied simultaneously when

$$A = B = \frac{\pi}{4}; C = \frac{\pi}{2}$$

$\Rightarrow$ Triangle is right angled isosceles.

Q.18 (A)

In $\triangle ABC$, we get

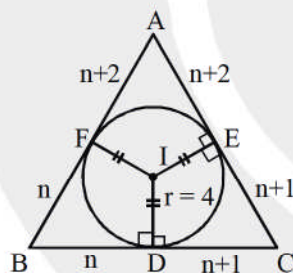
$$a^4 + b^4 + c^4 = b^2c^2 + 2c^2a^2 + 2a^2b^2 \quad (\text{By changing sine of angles into sides})$$

$$\Rightarrow (a^2)^2 - 2(b^2 + c^2)a^2 + (b^4 + c^4 - b^2c^2) = 0$$

$$\therefore a^2 = \frac{2(b^2 + c^2) \pm 2\sqrt{3}bc}{2} \Rightarrow a^2 = (b^2 + c^2) \pm \sqrt{3}bc \Rightarrow \left(\frac{b^2 + c^2 - a^2}{2bc} \right) = \pm \frac{\sqrt{3}}{2}$$

$$\therefore \cos A = \pm \frac{\sqrt{3}}{2} \Rightarrow A = 30^\circ \text{ or } 150^\circ \text{ Ans.}$$

Q.19 (C)



$$\therefore S = \frac{(2n+1) + (2n+3) + (2n+2)}{2} = 3n+3 = 3(n+1)$$

$$\text{Now, } r = \frac{\Delta}{s}$$

$$\Rightarrow 4 = \frac{\sqrt{s(s-a)(s-b)(s-c)}}{s} \Rightarrow 4 = \frac{\sqrt{3(n+1)(n+2)(n)(n+1)}}{3(n+1)} \Rightarrow 4 = \frac{\sqrt{3 \cdot n(n+2)}}{3}$$

$$\Rightarrow 144 = 3n(n+2) \Rightarrow 48 = n^2 + 2n \Rightarrow n^2 + 2n - 48 = 0 \Rightarrow (n+8)(n-6) = 0$$

$$\therefore n = 6$$

$$\text{So, } s = 3(n+1) = 3(6+1) = 3 \times 7 = 21 \text{ Ans.}$$

Q.20 (B)

A, B, C (angles of triangle are in A.P.)

$$\Rightarrow B = 60^\circ. \text{ So, } A + C = 120^\circ$$

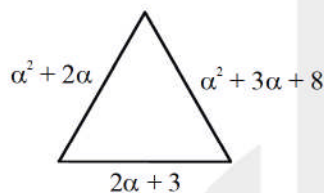
$$\text{Now, } \frac{b}{c} = \frac{\sqrt{3}}{\sqrt{2}} \Rightarrow \frac{\sin B}{\sin C} = \frac{\sqrt{3}}{\sqrt{2}} \Rightarrow \frac{\frac{\sqrt{3}}{2}}{\sin C} = \frac{\sqrt{3}}{\sqrt{2}} \Rightarrow \sin C = \frac{1}{\sqrt{2}}$$

$$\therefore C = 45^\circ \text{ or } 35^\circ$$

$$\text{But } C \neq 135^\circ$$

$$\text{So, } C = 45^\circ; A = 75^\circ$$

Q.21 (D)



$$\text{Now, } \alpha^2 + 2\alpha > 0 \Rightarrow \alpha(\alpha + 2) > 0 \Rightarrow \alpha > 0 \text{ or } \alpha < -2 \quad \dots\dots(1)$$

$$\text{Also, } 2\alpha + 3 > 0 \Rightarrow \alpha > \frac{-3}{2} \quad \dots\dots(2)$$

$$\text{Also, } \alpha^2 + 3\alpha + 8 > 0 \quad \forall \alpha \in \mathbb{R} \quad \dots\dots(3)$$

$$\therefore (1) \cap (2) \cap (3) \Rightarrow \alpha > 0$$

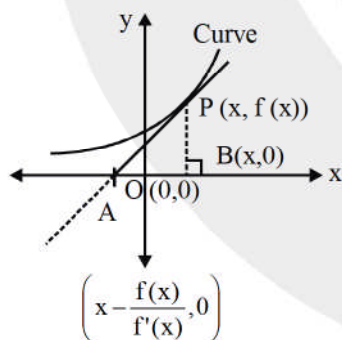
$$\text{Further, } \alpha^2 + 4\alpha + 3 > \alpha^2 + 3\alpha + 8 \Rightarrow \alpha > 5 \quad \dots\dots(4)$$

$$\alpha^2 + 5\alpha + 8 > 2\alpha + 3 \Rightarrow \alpha^2 + 3\alpha + 5 > 0 \quad \forall \alpha \in \mathbb{R} \quad \dots\dots(5)$$

$$\text{Also, } \alpha^2 + 5\alpha + 11 > \alpha^2 + 2\alpha \Rightarrow \alpha > -\frac{11}{3}$$

$\therefore$ Finally, we conclude that $\alpha > 5$ **Ans.**

Q.22 (A)



The equation of tangent at $P(x, f(x))$ is $(Y - f(x)) = f'(x)(X - x)$

$$\text{Put } Y = 0; A\left(X = x - \frac{f(x)}{f'(x)}, 0\right); B(x, 0)$$

$$\text{Now, } 2x - \frac{f(x)}{f'(x)} = 4 \Rightarrow 2(x - 2) = \frac{f(x)}{f'(x)} \Rightarrow \frac{f'(x)}{f(x)} = \frac{1}{2(x - 2)}$$

$$\Rightarrow 2 \ln|f(x)| = \ln|x - 2| + \ln c \Rightarrow y^2 = c(x - 2) \text{ Ans.}$$